

Lecture Notes: Histograms and Distributions

I. Histograms, Clearly Explained

What is a Histogram?

Imagine we measure the height of many people. If we plot each measurement as a dot, they overlap, and many are hidden.

We could try stacking dots that are *exactly* the same, but this is rare and doesn't solve the problem.

The Solution:

1. We **divide the range of values** (e.g., all the measured heights) into equal-sized intervals called **bins**.
2. We **stack the measurements** that fall into the same bin.
3. This resulting stack plot is a **histogram**.

The **taller** the stack in a bin, the **more measurements** we made that fall into that range.

What Do Histograms Tell Us?

Histograms allow us to see the **shape** of our data and **predict future probabilities**.

- We can bet that the next measurement will likely fall into one of the **tall bins**.
- Measurements in the **short bins** are rarer and less likely to happen.
- A histogram helps us **justify choosing a probability distribution** to approximate our data. For example, if the histogram is bell-shaped, we might be justified in using a **Normal Distribution**.
- If the data looked different (e.g., high on the left, long tail to the right), we might use an **Exponential Distribution**.

The Problem of Bin Width

Figuring out how wide to make the bins is a critical (and tricky) part of making a good histogram.

- **Bins are too narrow:** If every measurement gets its own bin, the histogram is just a flat, spiky, and useless plot. It gives no new insight.
- **Bins are too wide:** If the bins are so wide that all data falls into just one or two bins, we lose all the detail (e.g., just "above average" and "below average").

Rule of thumb: Don't trust the default settings in your software! You must **try a bunch of different bin widths** to find the one that gives you the clearest picture of your data's shape.

II. The Main Ideas Behind Probability Distributions

What is a Distribution?

Let's continue with our height measurements. When we put these measurements into bins, we get a **histogram**.

- The histogram shows us how the data is **distributed**.
- It gives us a sense of how **likely** we are to measure someone really tall, really short, or close to the average.

From Histogram to Distribution Curve

What if we measure *more* people and use *smaller* bin sizes?

- Our histogram becomes more **accurate** and **precise**.
- We can see more detail, like that most people are between 5.25 and 5.75 feet tall.

We can then draw a **smooth curve** to *approximate* this histogram. This curve is a **Probability Distribution**.

Advantages of the Curve (Distribution)

This curve tells us the same story as the histogram (low probability in the tails, high probability in the middle) but with a few key advantages:

1. **It fills in the gaps:** Even with many measurements, we might have empty bins just by random chance. The histogram would show a probability of 0 for that bin, but the curve correctly shows us there is still a *chance* of measuring a value in that range.
2. **It's not limited by bin width:** What's the probability of measuring someone between 5.021 and 5.317 feet? A histogram can't tell you (it's all in one bin). With the curve, we can use calculus (or a computer) to find the *exact* area under the curve between those two points.
3. **It saves time and money:** If we can't collect a ton of data, we can use the **mean** and **standard deviation** from our small sample to *approximate* the curve. This is often "good enough" and much cheaper.

Both the histogram and the curve are **distributions**. They show us how the probabilities of measurements are distributed.

III. The Normal Distribution, Clearly Explained

What is the Normal Distribution?

Chances are you've seen a **Normal** (or **Gaussian**) distribution before. It's also called a **bell-shaped curve** because it's a symmetrical curve that looks like a bell.

- **Example:** Human height.
- The **X-axis** represents the measurement (e.g., height in inches).
- The **Y-axis** represents the **relative probability** of observing that measurement.
 - It's **rare** to see someone super short, so the curve is **low** on the left.
 - It's **common** to see someone near the average, so the curve is **tall** in the middle.
 - It's **rare** to see someone super tall, so the curve is **low** on the right.

Parameters of the Normal Distribution

To draw any Normal Distribution, you only need to know two things (parameters):

1. The Mean (Average): μ

- This tells you the **center** of the curve.
- Normal distributions are always centered on their average value.
- **Example:** The average baby height is 20 inches, while the average adult height is 70 inches. The curves will be centered at 20 and 70, respectively.

2. The Standard Deviation: σ

- This tells you the **width** (or spread) of the curve.

- A **small** σ (like for babies) means the data is tightly clustered. This results in a **narrow** and **tall** curve.
- A **large** σ (like for adults) means the data is very spread out. This results in a **wide** and **short** curve.
- **Why the height difference?** The total area under *any* probability distribution must equal 1 (or 100%).
 - For babies, that 100% is squeezed into a narrow range, so the curve must be tall.
 - For adults, that same 100% is spread out over a wide range, so the curve must be short.

The 68-95-99.7 Rule

The standard deviation is incredibly useful because of a property of *all* Normal curves:

- **~95%** of all measurements fall between **plus or minus 2 standard deviations** ($\pm 2\sigma$) from the mean.
 - (The full rule is 68% for 1σ , 95% for 2σ , and 99.7% for 3σ).

Example:

- **Babies:** Mean = 20", $\sigma = 0.6$ ".
 - $2\sigma = 1.2$ ".
 - This means 95% of babies will be between (20 - 1.2) and (20 + 1.2), or **18.8" and 21.2"**.
- **Adults:** Mean = 70", $\sigma = 4$ ".
 - $2\sigma = 8$ ".
 - This means 95% of adults will be between (70 - 8) and (70 + 8), or **62" and 78"**.

Why is the Normal Distribution So Important?

It's magical how often we see the Normal distribution in nature (height, weight, commuting times, etc.). This isn't a coincidence.

The reason is a powerful statistical concept called the **Central Limit Theorem** (spoiler alert!). This theorem makes the Normal distribution super useful for all kinds of statistics.

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